

University of Milan-Bicocca

Department of Statistics and Quantitative Methods



Populism in Power and Political Corruption

Association, Temporal Ordering, and Institutional
Decomposition

Data Science Laboratory Project

Authors

Irida Pajenga

`i.pajenga@campus.unimib.it`

Vladyslav Yelisieiev

`v.yelisieiev@campus.unimib.it`

Simone Ragusini

`s.ragusini@campus.unimib.it`

Edoardo Zero

`e.zero1@campus.unimib.it`

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Abstract

Populist parties often campaign against corrupt elites, yet governments that concentrate political power may also weaken oversight and facilitate institutional capture. We examine this tension using the V-Dem Country-Year v16 and V-Party V2 datasets. Party-level expert assessments are aggregated to the senior governing party, filtered for coder reliability, expanded between elections, and joined to annual country data. The resulting unbalanced panel contains 3,965 country-years for 96 countries between 1970 and 2019.

The analysis separates three questions that are frequently conflated: contemporaneous association, temporal ordering, and institutional heterogeneity. First, two-way fixed-effects models compare each country with itself while absorbing common year shocks and controlling for economic development and electoral democracy. Second, bidirectional lead-lag and Granger-style specifications, placebo leads, event studies, first differences, and election-based samples probe whether populism precedes corruption or vice versa. Third, the composite corruption index is decomposed into executive, public-sector, legislative, and judicial dimensions, with false-discovery-rate correction.

The positive pooled association between populist government and corruption is imprecise after panel dependence is considered and reverses sign within countries. The controlled same-year estimate is small and negative, statistically significant in the main specification but not under every robustness design. We find no robust temporal direction for the composite index. Decomposition shows that the negative contemporaneous association is strongest for judicial corruption. The only positive temporal signal is a small two-year populism-to-legislative-corruption coefficient; it survives the primary Granger and multiplicity tests but not first-difference or election-only checks. Overall, the data do not support a broad claim that populist government systematically raises political corruption. They instead reveal a nuanced, institution-specific, and fragile pattern that warrants cautious interpretation rather than a causal conclusion.

Keywords: populism; political corruption; panel data; fixed effects; Granger-style tests; V-Dem; V-Party; institutional capture.

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Introduction

Populism presents a genuine empirical puzzle for the study of corruption. Populist parties define politics as a moral conflict between a virtuous people and a self-serving elite (Mudde, 2004). This framing makes anti-corruption rhetoric electorally natural: established parties, bureaucracies, courts, and media can all be portrayed as parts of a captured system. Yet the same majoritarian logic may weaken horizontal accountability once populists enter government. Concentrating executive authority, politicising appointments, and discrediting independent checks can create opportunities for institutional capture (Chesterley and Roberti, 2018; Batory, 2016). The relationship can also run in reverse. Persistent corruption and declining trust may increase demand for outsiders who promise to displace established elites (Gurieva and Papaioannou, 2022; Kubbe and Loli, 2020).

These mechanisms imply that a simple cross-country correlation is not enough. Countries in which populists govern differ systematically in economic development, democratic institutions, regional history, and baseline corruption. Moreover, both governing-party ideology and corruption change slowly. A positive pooled relationship can therefore reflect stable differences between countries, while a same-year fixed-effects estimate cannot establish which variable came first. Finally, an aggregate corruption index may hide divergent effects across the executive, legislature, judiciary, and public administration. Existing research likewise emphasises that populism is heterogeneous and its institutional effects need not be identical across ideological or political settings (Huber and Schimpf, 2017; Engler, 2024).

This report addresses these problems through three research questions:

- RQ1.** Within a country, are years of more populist government associated with higher political corruption after accounting for country baselines, common year shocks, income, and democracy?
- RQ2.** Does governing-party populism predict subsequent corruption, does corruption predict subsequent populism, or is there no robust temporal ordering?
- RQ3.** Do the results differ across executive, public-sector, legislative, and judicial corruption?

The empirical contribution is not a search for a model that produces the strongest result. Instead, each method answers a distinct question. Pooled and between-country estimates describe cross-national differences. Two-way fixed effects isolate within-country co-movement. Granger-style models ask whether one variable adds predictive information beyond the other's own persistence. Event studies, placebo leads, first differences, alternative electoral samples, and multiple-testing correction delimit the claims that can be sustained.

The findings are deliberately nuanced. The pooled populism coefficient is positive, but statistically imprecise once country dependence is recognised. Within countries, the coefficient becomes small and negative. It is significant in the controlled static model, but its significance is sensitive to alternative measurement and differencing choices. The composite index provides no robust evidence for either temporal direction. Decomposition clarifies that judicial corruption accounts for the strongest negative same-year pattern, whereas a small positive two-year legislative signal is the only dimension-specific temporal exception. That signal survives the primary Granger

and false-discovery tests but disappears in first-difference and election-only designs.

The central conclusion is therefore not that populism reduces corruption, nor that corruption is irrelevant. Rather, the headline cross-country association does not translate into a general within-country or temporal effect. The evidence is more consistent with strong contextual confounding, slow-moving measurement, and limited institution-specific dynamics. This negative result is substantively useful: it identifies where a widely assumed relationship does and does not survive a more demanding panel design.

The remainder of the report proceeds as follows. [Chapter 2](#) describes the data, measurement, sample construction, and exploratory evidence. [Chapter 3](#) sets out the estimands and model specifications. [Chapter 4](#) presents the association, temporal, and decomposition findings, then discusses limitations and implications. The appendix documents robustness and reproducibility.

Data and Descriptive Evidence

2.1 Data sources

The analysis combines two datasets from the Varieties of Democracy project. The V-Dem Country-Year Dataset v16 provides annual political indicators, including corruption, electoral democracy, income, population, and region (Coppedge et al., 2026). The V-Party V2 dataset provides party-level expert assessments of ideology and organisation at election dates, including populism and government status (Lindberg et al., 2022). V-Dem’s measurement model aggregates multiple expert ratings into comparable latent estimates while explicitly modelling coder reliability and uncertainty (Pemstein et al., 2026). These data offer unusually broad temporal and geographic coverage, but they remain expert-coded measures rather than direct administrative records.

The two sources have different units of observation. V-Dem is a country-year panel. V-Party is a country-party-election dataset and is empirically dense only in election years. Linking them therefore requires substantive choices about which party represents the government and how an election-year score should be carried into subsequent years. Figure 2.1 summarises the complete workflow.

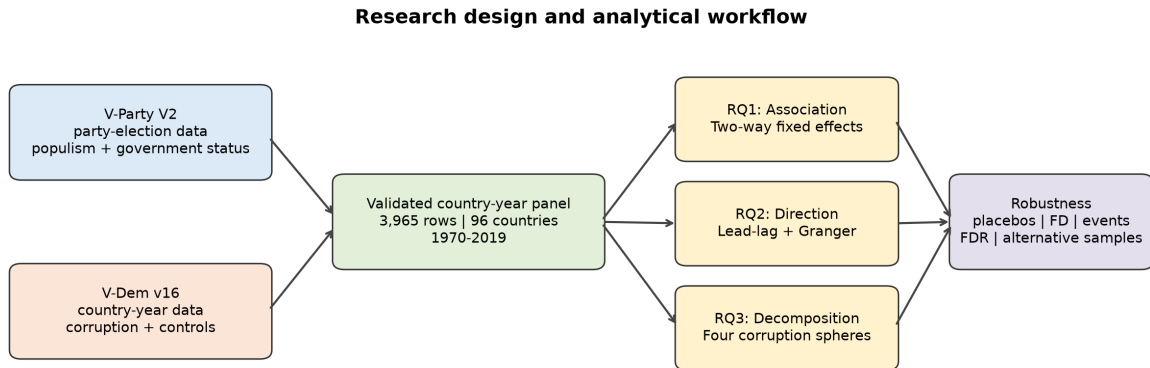


Figure 2.1: Research design and analytical workflow. Party-election populism scores are filtered, aggregated, forward-filled, and joined to annual V-Dem outcomes before the three research questions are estimated.

2.2 Core measures

2.2.1 Governing-party populism

The primary explanatory variable is V-Party’s `v2xpa_popul`, a continuous index from 0 to 1 constructed from people-centrism and anti-elitism. This operationalisation matches the ideational definition of populism as a thin-centred opposition between the people and the elite (Mudde, 2004). The index measures rhetoric and party identity; it should not be read as a

complete measure of nationalism, policy radicalism, or democratic quality.

The analysis focuses on parties coded as the senior governing partner (`v2pagovsup` = 0), normally the prime minister’s party. If a country-election has more than one senior governing party, their populism scores are averaged using vote shares when all weights are available and a simple mean otherwise. Populism is modelled continuously. The value 0.47, the 75th percentile in the raw governing-party distribution, is used only for descriptive labels and never as a regression threshold.

2.2.2 Political corruption

The primary outcome, `v2x_corr`, is V-Dem’s composite political corruption index. It ranges from 0 (cleaner) to 1 (more corrupt) and averages corruption in four institutional spheres. The decomposition uses executive corruption (`v2x_execorr`), public-sector corruption (`v2x_pubcorr`), legislative corruption (`v2lgcrrpt`), and judicial corruption (`v2jucorrdc`). The last two continuous latent estimates use the opposite source orientation: higher values indicate cleaner institutions. They are multiplied by -1 in decomposition models so that a positive coefficient always means more corruption.

The project also constructs five-level 0–1 versions of legislative and judicial corruption from the ordinal ratings. These harmonised variables are useful for plots but are too coarse to be the primary fixed-effects outcomes: several countries never change ordinal category. The continuous latent estimates preserve substantially more within-country information.

Table 2.1: Variables used in the empirical analysis.

Role	Variable	Interpretation	Model use
Treatment	<code>populism_governing</code>	Senior governing party’s continuous populism score, forward-filled between elections	All primary models
Composite outcome	<code>v2x_corr</code>	Political corruption, 0 cleaner to 1 more corrupt	RQ1–RQ2
Executive outcome	<code>v2x_execorr</code>	Bribery and embezzlement by the executive	RQ3
Public-sector outcome	<code>v2x_pubcorr</code>	Bribery and theft in public administration	RQ3
Legislative outcome	<code>v2lgcrrpt</code>	Continuous legislative corruption estimate; sign reversed in models	RQ3
Judicial outcome	<code>v2jucorrdc</code>	Continuous judicial corruption estimate; sign reversed in models	RQ3
Economic control	<code>log_gdppc</code>	Natural logarithm of GDP per capita	FE and direction models
Political control	<code>v2x_polyarchy</code>	Electoral democracy, 0 to 1	FE and direction models
Context	<code>e_regiongeo</code> , <code>e_pop</code>	Region and population	Description / CV only

Region is time-invariant and therefore absorbed by country fixed effects. Population is not an estimated control. This avoids presenting columns retained for description as if they entered the reported models.

2.3 Panel construction

The analytical window is 1970–2019. V-Party contains records before 1970, but its expert-coded populism and government-status variables are unpopulated in those earlier years. Coverage ends in 2019, so the analysis does not include the most recent populist-government episodes.

The country-year panel is built in five steps:

1. Restrict V-Party to senior governing parties and the 1970–2019 window.
2. Require more than three coders for both expert-coded components of the populism index. The composite index itself has no coder-count field, so filtering its components is the appropriate quality check.
3. Aggregate multiple senior governing parties to one country-election observation.
4. Forward-fill the election observation until the next coded election, without filling years before a country’s first observation.
5. Inner-join the result to V-Dem on `country_id` and `year`; validate unique, gap-free annual histories before constructing lags and leads.

Forward-filling is necessary to align election-dated party information with annual corruption data, but it is also an assumption: governing-party populism is treated as constant between elections. The models therefore contain many annual rows but fewer genuinely distinct changes in government populism. Election-year and government-change-year estimates are reported as robustness checks.

Table 2.2: Sample construction and effective estimation samples.

Stage	Observations	Countries	Interpretation
Senior governing-party observations before strict coder filter	1,628	163	Election-dated source
After component coder-count filter	953	96	Reliable election observations
Final annual panel after forward-fill and join	3,965	96	1970–2019 panel
M1–M3 complete-case stability sample	3,940	95	Main association sample
M4 sample with lagged corruption	3,845	95	Dynamic short-run model
Two-lag composite direction sample	3,750	95	Main lead-lag / Granger sample
Legislative outcome available	3,770	95	Structurally lower coverage

The strict coder filter trades representation for reliability. It removes 67 countries present before filtering, and the resulting estimates should not be generalised automatically to countries with sparser expert coverage. At the same time, the panel retains 95–96 countries and up to 50 annual observations per country, which is substantial for a within-country design.

2.4 Exploratory evidence

2.4.1 Distribution and trends

In the broader election-year source, governing-party populism is unimodal and right-skewed (mean 0.330, median 0.274, 75th percentile 0.470). There is no empirical break that would justify turning the index into a populist/non-populist dummy. The global mean drifts upward from roughly 0.30 in the 1970s to 0.36 in the 2010s, but the pattern is gradual and noisy rather than a single post-2010 break.

Composite corruption is bimodal, with one cluster of clean democracies near zero and another of high-corruption regimes near 0.85. Regional differences are much larger than the descriptive high-versus-lower-populism gap: mean corruption ranges from approximately 0.05 in Western Europe and North America to approximately 0.85 in Central Asia. These differences motivate the separation of between-country and within-country variation.

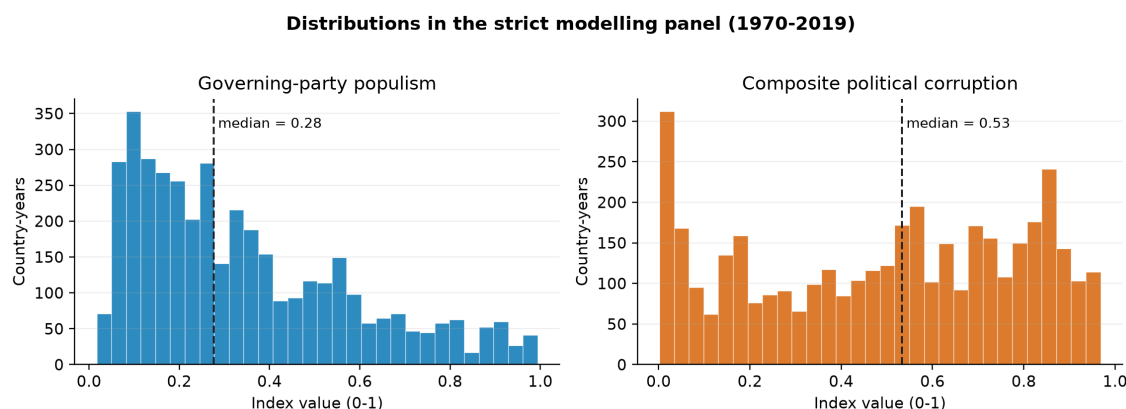


Figure 2.2: Distributions of the treatment and composite outcome in the strict modelling panel. Dashed lines mark medians. These histograms describe the estimation sample and are not evidence of a causal relationship.

2.4.2 Descriptive group contrasts

For presentation only, the exploratory notebook labels the top quartile of governing-party populism as “high populism.” Across 1,831 high-populism and 5,078 lower-populism country-years in the broader descriptive panel, all corruption measures are higher in the high-populism group. Yet Cliff’s delta is small for every dimension (approximately 0.05–0.11). Legislative and judicial corruption show the largest gaps, whereas executive and public-sector corruption show the smallest. Large sample size makes the group differences statistically detectable; it does not make them substantively large.

This descriptive result is vulnerable to confounding because it compares different countries and regions. It is therefore treated as motivation for RQ1 and RQ3, not as a finding about what happens after a country elects a more populist government.

2.4.3 Illustrative governing transitions

Five case studies reinforce the ambiguity. Comparing five-year average composite corruption before and after a documented populist entry, Venezuela increases from 0.69 to 0.87 and Hungary from 0.28 to 0.40. Turkey changes only slightly (0.50 to 0.53), Bolivia is essentially flat (0.70 to 0.70), and Ecuador declines slightly (0.71 to 0.67). Venezuela, Bolivia, and Ecuador were already relatively corrupt before populist entry, whereas Hungary begins from a cleaner baseline and more closely resembles an institutional-capture narrative.

The cases demonstrate why neither a before/after anecdote nor a cross-sectional correlation can resolve direction. Populism may precede corruption, corruption may create demand for populism, or both may reflect common institutional and economic shocks. The next chapter translates these alternatives into distinct panel estimands.

2.5 Measurement and data-quality risks

Four limitations are carried into the empirical design. First, V-Dem and V-Party are both expert-coded, so shared perceptions can induce correlated measurement error. Second, forward-filled populism changes mainly around elections, while corruption is extremely persistent. Third, legislative corruption is missing when no legislature exists, making missingness institutionally meaningful rather than random. Fourth, the five-level legislative and judicial ordinal measures are often constant within countries. The report therefore uses continuous point estimates for primary models, clustered inference, first-difference checks, election-based samples, and explicit diagnostics of within-country variation.

Empirical Strategy

3.1 Estimands before models

The phrase “populism is associated with corruption” can describe three different quantities. A pooled estimate combines differences between countries and changes within countries. A between estimate asks whether countries with more populist governments are more corrupt on average. A fixed-effects estimate asks whether a country is more corrupt in years when its own government is more populist than usual. Finally, a directional model asks whether past values of one variable predict later values of the other after persistence is considered.

No single model answers all these questions. The analysis therefore fixes the estimand first and uses model fit only as a diagnostic. In particular, a pooled regression may predict levels well because it uses large cross-country differences, yet remain inappropriate for the within-country research question. Conversely, a low within- R^2 does not invalidate a fixed-effects coefficient when the dependent variable is highly persistent and the purpose is estimation rather than prediction (Wooldridge, 2010).

3.2 Contemporaneous association

3.2.1 Two-way fixed effects

The primary association model is

$$C_{it} = \beta P_{it} + \mathbf{X}_{it}'\boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (3.1)$$

where C_{it} is composite corruption, P_{it} is governing-party populism, $\mathbf{X}_{it}$ contains time-varying controls, α_i are country fixed effects, and λ_t are year fixed effects. Country effects remove time-invariant history, geography, and baseline institutions. Year effects remove shocks common to all countries. The coefficient β is therefore identified by within-country deviations from long-run country and year means.

Three static specifications are fitted on one frozen complete-case sample:

- M1: populism plus country and year effects;
- M2: M1 plus log GDP per capita;
- M3: M2 plus electoral democracy.

Freezing the M1–M3 sample ensures that coefficient movement reflects added controls rather than a changing set of observations. Country-clustered standard errors allow arbitrary serial dependence within countries. Two-way clustering and Driscoll–Kraay covariance estimates provide sensitivity to common cross-sectional shocks (Driscoll and Kraay, 1998).

M4 adds lagged corruption:

$$C_{it} = \rho C_{i,t-1} + \beta_{SR} P_{it} + \mathbf{X}_{it}' \boldsymbol{\gamma} + \alpha_i + \lambda_t + u_{it}. \quad (3.2)$$

This is not a fourth stability model. It asks about the small short-run residual change after last year's corruption is considered. Its coefficient and within- R^2 are not directly comparable to static M1–M3. Dynamic fixed-effects estimates may also exhibit finite-panel bias (Nickell, 1981); the relatively long country histories reduce, but do not remove, that concern.

3.2.2 Between-versus-within comparison

To locate the raw positive relationship, the same single-predictor association is estimated as pooled OLS, between-country OLS on country means, country fixed effects, and two-way fixed effects. Pooled errors are clustered by country and between-country errors are heteroskedasticity-robust. The purpose is not to select a winner by fit, but to show whether the sign comes from comparing different countries or from changes within the same country.

3.2.3 Democracy heterogeneity

The hypothesis that institutional context moderates the association is tested by interacting populism with electoral democracy:

$$C_{it} = \beta_1 P_{it} + \beta_2 D_{it} + \beta_3 (P_{it} D_{it}) + \gamma \log GDPpc_{it} + \alpha_i + \lambda_t + \varepsilon_{it}. \quad (3.3)$$

Rather than interpret β_3 alone, the report presents the model-implied slope $\partial C / \partial P = \beta_1 + \beta_3 D$ at the 10th, 50th, and 90th percentiles of democracy. These are marginal effects from one interaction model, not separate regressions chosen after observing the data.

3.3 Temporal ordering

3.3.1 Lead-lag models

RQ2 tests both possible directions. For populism to corruption, the two-lag model is

$$C_{it} = \theta_1 P_{i,t-1} + \theta_2 P_{i,t-2} + \mathbf{X}_{it}' \boldsymbol{\gamma} + \alpha_i + \lambda_t + e_{it}. \quad (3.4)$$

The reverse model exchanges C and P . Lead-lag estimates establish temporal order more clearly than same-year association, but remain vulnerable to persistence. In the analysis panel, lag-1 autocorrelation is approximately 0.93 for populism and 0.99 for composite corruption.

3.3.2 Granger-style tests

The stricter model conditions on the dependent variable's own history:

$$C_{it} = \sum_{\ell=1}^2 \rho_{\ell} C_{i,t-\ell} + \sum_{\ell=1}^2 \theta_{\ell} P_{i,t-\ell} + \mathbf{X}_{it}' \boldsymbol{\gamma} + \alpha_i + \lambda_t + e_{it}. \quad (3.5)$$

A joint Wald test evaluates $H_0 : \theta_1 = \theta_2 = 0$. Rejection means that lagged populism adds predictive information beyond corruption’s own past. It does not establish structural causality: unobserved time-varying confounders, measurement error, and anticipation can still generate predictive ordering. We therefore use “Granger-style” rather than causal language ([Granger, 1969](#)).

Lag windows from one to five years are compared with AIC and BIC only as diagnostics. Longer windows drop additional early observations, so the information criteria do not always compare identical samples. The two-lag model remains the pre-specified main analysis.

3.3.3 Placebos, events, and changes

Three designs probe the main directional result:

1. **Placebo leads.** Future proposed-cause values are added to the current-outcome model. Significant future terms would warn of pre-trends, anticipation, or omitted common causes.
2. **Event studies.** Populist entry is the first election-year increase in governing populism of at least 0.10 that reaches 0.50. A corruption shock is the first annual increase in the composite index of at least 0.05. Event-time coefficients are estimated relative to year -1 , with joint pre- and post-event tests.
3. **First differences.** Annual changes remove country levels through an alternative transformation. Directional models regress the current change in the outcome on the lagged change in the proposed cause and changes in controls.

Because populism is forward-filled, the lead-lag model is also re-estimated on election years, government-change years, and observations no more than five or ten years from the last election.

3.4 Institutional decomposition

For each corruption dimension d , the controlled static model is

$$C_{it}^d = \beta_d P_{it} + \mathbf{X}_{it}' \boldsymbol{\gamma}_d + \alpha_i^d + \lambda_t^d + \varepsilon_{it}^d. \quad (3.6)$$

Raw coefficients cannot be ranked across outcomes with different latent scales. We therefore report the within-standardised effect

$$\tilde{\beta}_d = \beta_d \frac{SD_{within}(P)}{SD_{within}(C^d)}. \quad (3.7)$$

This quantity is the change in within-country standard deviations of the outcome associated with a one-within-standard-deviation change in populism. Outcome-specific samples preserve power; a common complete-case sample checks whether rankings reflect legislative missingness.

The primary decomposition contains four same-year M3 tests and eight Granger-style tests (four dimensions in two directions). Benjamini–Hochberg adjustment controls the false discovery rate within these pre-declared families ([Benjamini and Hochberg, 1995](#)). The composite is a benchmark and is not counted as an independent fifth corruption outcome.

3.5 Predictive benchmark

A gradient-boosted tree model provides a separate non-linear diagnostic. XGBoost uses governing-party populism, log GDP per capita, and electoral democracy ([Chen and Guestrin, 2016](#)). Entire countries are held out with region-stratified group folds, preventing observations from the same country appearing in both training and test sets. A nested inner loop tunes the hyperparameters without seeing outer-fold test countries. SHAP values rank feature use in the fitted prediction model ([Lundberg and Lee, 2017](#)).

This exercise is not an alternative causal estimator. Raw features retain between-country information that fixed effects deliberately remove, so out-of-fold predictive R^2 and fixed-effects within- R^2 answer different questions. The useful diagnostic is whether a flexible model finds populism important after GDP and democracy are available.

3.6 Decision rules and reproducibility

The analysis follows four reporting rules. First, coefficients, intervals, sample sizes, and effect magnitudes are reported together; statistical significance is never treated as substantive size. Second, null findings are stated as absence of robust evidence, not proof of no effect. Third, robustness checks evaluate stability and do not form a leaderboard from which the preferred result is selected. Fourth, causal wording is reserved for designs that could justify it; this observational panel cannot.

All preprocessing and analysis steps are executable from versioned Python modules. Tables and figures are generated from committed processed data, and automated tests validate aggregation, orientation, panel shifts, model construction, and multiplicity correction. Reproducibility commands are listed in [Section A.4](#).

Results and Discussion

4.1 Where does the positive descriptive association come from?

The exploratory comparison suggests that country-years governed by highly populist parties are more corrupt. The model comparison shows why that pattern should not be interpreted as a within-country effect. Pooled OLS gives a positive populism coefficient of 0.145 (95% CI $[-0.024, 0.315]$, $p = 0.093$). The between-country estimate is larger at 0.226, but also imprecise (95% CI $[-0.065, 0.516]$, $p = 0.127$). Once country effects are introduced, the sign flips: the country-FE estimate is -0.065 and the two-way FE estimate is -0.068 . Neither single-predictor within estimate is significant at 5%, but their signs are stable.

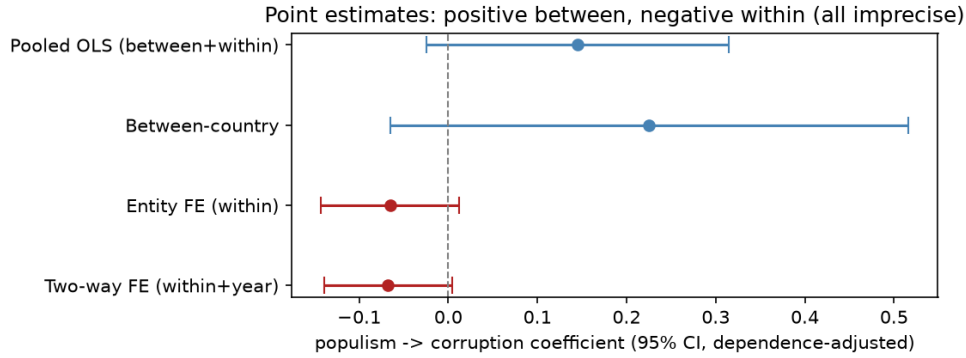


Figure 4.1: Between-versus-within comparison. Positive pooled and between-country point estimates become negative in country fixed-effects models; all four single-predictor estimates are imprecise after dependence-adjusted inference.

The evidence therefore supports a careful formulation: countries with more populist governments tend to have higher average corruption, but the estimate is uncertain and does not survive the shift to a within-country comparison. Stable regional and institutional differences plausibly account for much of the descriptive association.

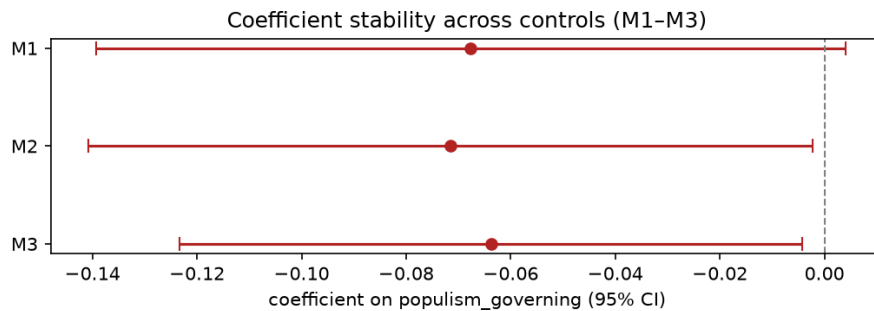
4.2 Composite corruption: controlled association

Table 4.1 presents the nested fixed-effects models. Adding log GDP per capita makes the negative coefficient marginally more precise; adding electoral democracy leaves it close to -0.064 . M2 and M3 exclude zero at the 5% level. The sign does not support the hypothesised positive same-year relationship.

Table 4.1: Two-way fixed-effects association models for composite corruption.

Model	Added terms	Populism coefficient	95% CI	p	Within- R^2
M1	Country + year FE	-0.0677	[-0.139, 0.004]	0.064	0.010
M2	+ log GDP per capita	-0.0715	[-0.141, -0.002]	0.043	-0.059
M3	+ electoral democracy	-0.0638	[-0.123, -0.004]	0.036	-0.196
M4	+ lagged corruption	-0.0033	[-0.016, 0.010]	0.615	0.885

M1–M3 use the same 3,940 observations and 95 countries. M4 uses 3,845 observations and answers a different short-run question after accounting for outcome persistence. Negative within- R^2 can occur because the reported metric compares the transformed fit with a benchmark; it does not reverse the coefficient interpretation.

**Figure 4.2:** Stability of the populism coefficient across the static M1–M3 specifications. Country-clustered 95% confidence intervals are shown.

The coefficient is statistically detectable in the main controlled model but substantively small. A one-within-country-standard-deviation increase in governing-party populism corresponds to an estimated 0.010 reduction in the 0–1 corruption index. This is approximately 3% of the overall corruption standard deviation, or 10% of its within-country standard deviation. A typical election-to-election change implies an even smaller movement.

Inference is stable under country-and-year clustering and Driscoll–Kraay standard errors. First differences remain negative (-0.024) but are not distinguishable from zero ($p = 0.195$). Alternative government aggregations and originally proposed institutional controls also retain the negative sign but reduce precision. The appropriate summary is therefore *sign-robust but not significance-robust*; it is not evidence that populism causes lower corruption.

4.2.1 Heterogeneity by electoral democracy

The populism-democracy interaction is statistically meaningful. The model-implied populism slope is -0.164 at the 10th percentile of polyarchy and -0.076 at the median, with confidence intervals excluding zero. At the 90th percentile it is $+0.027$ and imprecise. The negative same-year pattern is therefore concentrated in less democratic contexts. This could represent genuine institutional heterogeneity, but it could also reflect expert coding, regime trajectories, or conditioning on a variable partly affected by populist government. We treat it as descriptive moderation rather than a causal mechanism.

4.2.2 Non-linear predictive benchmark

The country-held-out XGBoost model obtains out-of-fold R^2 around 0.45, compared with 0.38 for a linear model on the same folds. Nested tuning increases XGBoost performance to approximately 0.51 and selects shallow, regularised trees. In both untuned and tuned fits, populism is the least important of the three predictors; electoral democracy and GDP dominate SHAP importance.

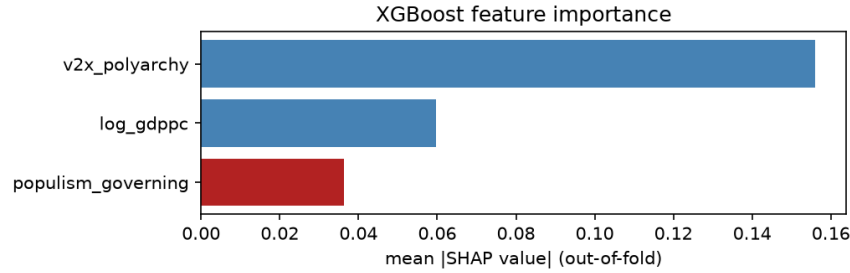


Figure 4.3: Out-of-fold XGBoost feature importance. SHAP magnitude describes model use, not statistical significance or causality.

This benchmark reinforces a limited conclusion: even a flexible model that is allowed to use between-country information relies relatively little on populism to predict corruption.

4.3 Temporal direction for composite corruption

The simple two-lag populism-to-corruption model gives a negative lag-1 coefficient of -0.048 ($p = 0.031$) and a near-zero lag-2 coefficient. This is opposite to the hypothesised positive delayed effect. In the reverse model, neither corruption lag is individually clear. Once the dependent variable's own history is included, the primary Granger-style joint tests do not reject either null:

- populism $\rightarrow$ corruption: joint $p = 0.686$;
- corruption $\rightarrow$ populism: joint $p = 0.241$.

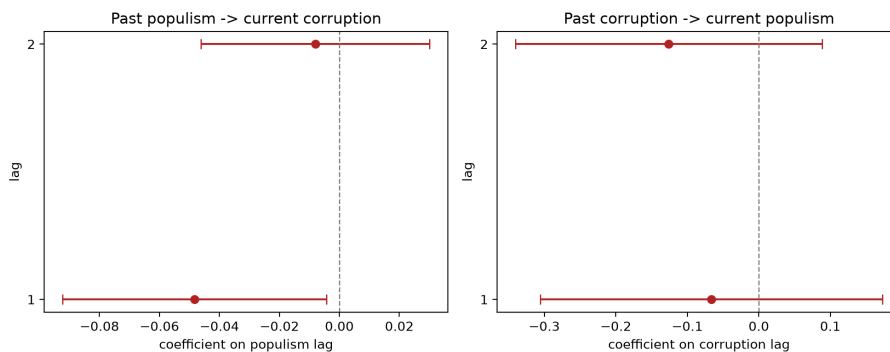


Figure 4.4: Controlled lead-lag coefficients in both proposed directions. The simple lag-1 populism-to-corruption estimate is negative, but the stricter joint Granger-style test is null.

Placebo leads do not provide a strong warning: future corruption terms jointly have $p = 0.841$ in the reverse model, and future populism terms have $p = 0.096$ in the forward model. First

differences are also null in both directions. AIC and BIC weakly favour one-year Granger windows, but longer-window comparisons use progressively smaller samples and are not treated as model-selection evidence.

4.3.1 Event-study evidence

The populist-entry event study identifies 45 events. Pre-event coefficients are jointly null ($p = 0.983$), which is consistent with no detectable pre-trend, but post-entry coefficients are also jointly null ($p = 0.228$). The estimates do not reveal a statistically detectable corruption increase after entry.

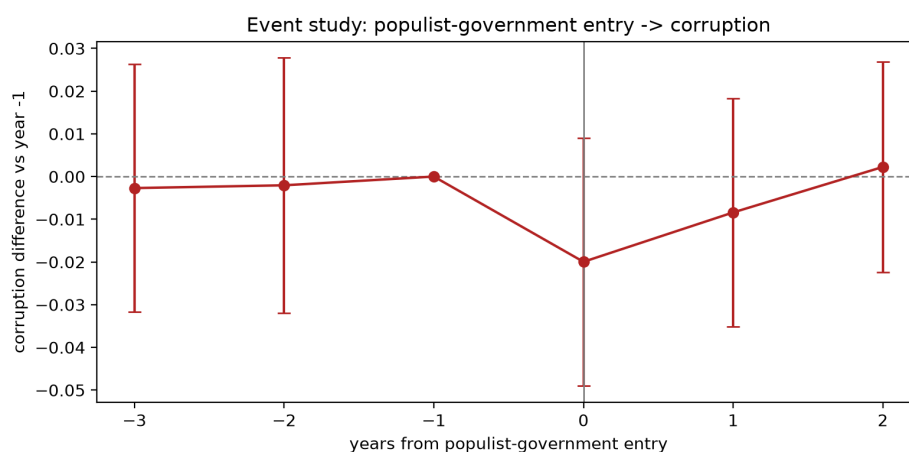


Figure 4.5: Composite corruption around populist entry. Event time -1 is the reference. Neither the pre-trend nor post-entry joint test is significant.

The reverse corruption-shock event study finds a post-event joint signal, but governing populism is already elevated before the shock and the pre-trend test is borderline. The sequence is therefore more consistent with political episodes in which both variables cluster than with a clean corruption-first mechanism.

Taken together, the directional evidence does not support a robust temporal effect for the composite index. The small negative same-year association should not be converted into a claim about temporal causation.

4.4 Institutional decomposition

The composite index may conceal relationships concentrated in one institution. [Table 4.2](#) reports within-standardised M3 coefficients and primary Granger-style joint tests. Positive values are oriented toward more corruption for every outcome, and FDR correction covers the four subtype association tests and eight subtype-by-direction Granger tests.

Table 4.2: Primary corruption-dimension results.

Outcome	Contemporaneous M3		Granger joint p		N_{M3}
	Std. β	FDR q	$P \rightarrow C$	$C \rightarrow P$	
Composite benchmark	-0.097	–	0.686	0.241	3,940
Executive	0.021	0.685	0.496	0.870	3,940
Public sector	-0.097	0.113	0.734	0.227	3,940
Legislative	-0.076	0.213	0.003	0.064	3,745
Judicial	-0.173	0.009	0.316	0.251	3,940

Granger cells show raw joint p values for compactness. The legislative forward-direction test has FDR $q = 0.028$; all other subtype direction tests have $q > 0.25$. The composite is a benchmark and is excluded from subtype FDR families.

4.4.1 Contemporaneous dimension results

The negative composite association is not uniform. Executive corruption is near zero ($\tilde{\beta} = 0.021$, $q = 0.685$). Public-sector and legislative corruption are negative but imprecise after FDR correction. Judicial corruption has the clearest negative association ($\tilde{\beta} = -0.173$, 95% CI $[-0.284, -0.063]$, raw $p = 0.002$, $q = 0.009$).

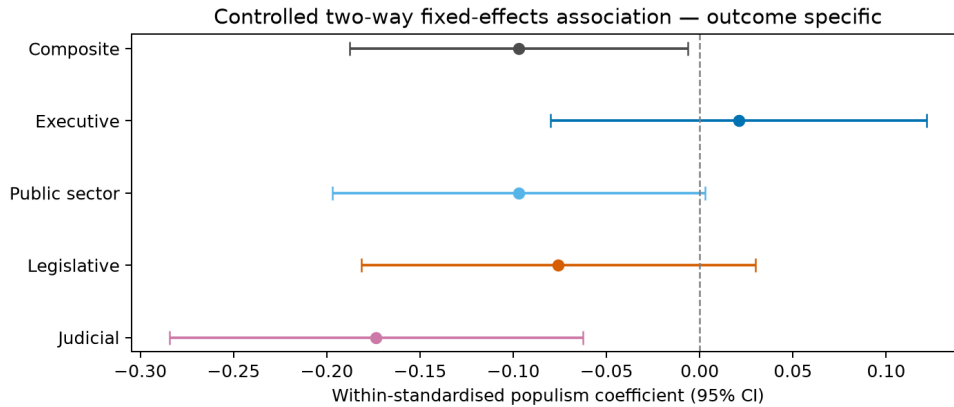


Figure 4.6: Controlled two-way fixed-effects association by corruption dimension. Effects are scaled by within-country standard deviations so that magnitudes are comparable.

The common complete-case sample preserves the ranking. The judicial estimate remains negative ($\tilde{\beta} = -0.144$, $q = 0.042$), so the result is not explained by its larger available sample. Its point estimate is unchanged under country, country-and-year, and Driscoll–Kraay covariance estimators. A contemporaneous first-difference model retains the negative sign but does not survive four-outcome correction (raw $p = 0.033$, $q = 0.134$). The coarse five-level judicial measure is also negative ($\tilde{\beta} = -0.137$, $p = 0.026$). The sign is therefore robust to measurement and inference choices, while confirmatory significance is weaker under differencing.

The judicial interaction mirrors the composite: the populism slope is negative at low and median polyarchy and approximately zero at high polyarchy. This secondary result is consistent with institutional heterogeneity, but it may also reflect low within-country resolution or expert coding in less democratic regimes.

4.4.2 Dimension-specific temporal ordering

No subtype Granger test is significant after FDR except populism to legislative corruption (raw joint $p = 0.0035$, $q = 0.0279$). The lag-1 coefficient is -0.0328 ($p = 0.236$); the lag-2 coefficient is $+0.0733$ (95% CI $[0.0234, 0.1231]$, $p = 0.004$). In within-standardised terms, the two-year effect is only $+0.0234$. The reverse legislative direction does not survive multiplicity correction ($p = 0.064$, $q = 0.257$).

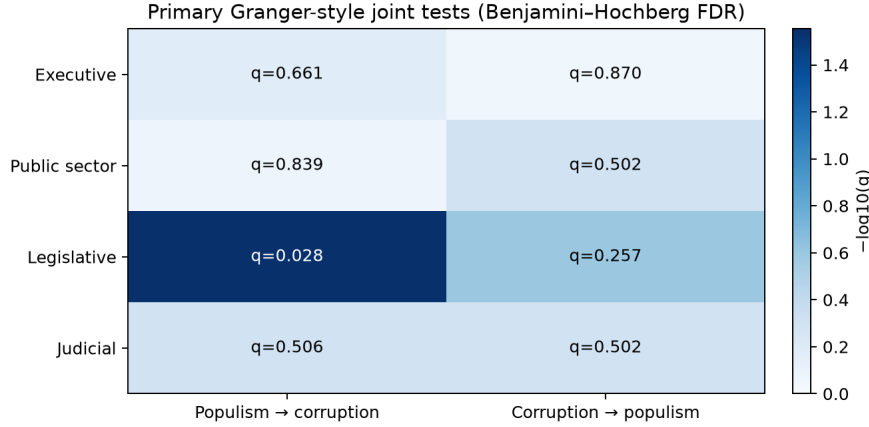


Figure 4.7: FDR-adjusted joint Granger-style tests across corruption dimensions and directions. Only the populism-to-legislative-corruption family falls below 0.05.

The legislative signal passes the common-sample and placebo-lead checks: future populism does not jointly predict current legislative corruption after FDR ($q = 0.239$). However, it is not confirmed by first differences (raw $p = 0.374$), election-year observations ($p = 0.701$), or government-change years ($p = 0.486$). Simpler lead-lag models retain a signal in the full and forward-fill-restricted samples.

The evidence is consequently specific but fragile. A positive two-year coefficient is compatible with slower legislative capture, but dependence on level-based, forward-filled information prevents a causal interpretation. The decomposition refines the composite null result; it does not overturn it.

4.5 Discussion

Three findings matter. First, the eye-catching positive cross-country relationship is an imprecise between-country pattern. It reverses sign in within-country models, indicating that stable differences between countries are central. This is consistent with the very large regional and regime-type gaps observed in the exploratory analysis.

Second, the composite panel does not reveal a robust temporal direction. Populism and corruption are both persistent and may respond to the same political shocks. The event and placebo designs do not support a general sequence in which populist entry is followed by rising corruption, nor one in which corruption reliably brings populists to office. This absence of evidence describes what these annual panel designs can detect; it is not proof that every causal effect is exactly zero.

Third, aggregation matters. Judicial corruption produces the strongest negative contemporane-

ous association, while legislative corruption produces the only positive delayed signal. Neither pattern justifies a simple causal story. The judicial result may capture coding and regime context, especially because it is strongest at low democracy. The legislative result is theoretically compatible with slower institutional capture (Chesterley and Roberti, 2018; Batory, 2016), but its failure in first-difference and election-only samples makes it provisional.

The results also caution against interpreting anti-corruption rhetoric as either proof of cleaner government or proof of hidden corruption. Populism is a broad political style and ideology whose institutional consequences vary with constraints, party organisation, and democratic context (Gurieva and Papaioannou, 2022; Huber and Schimpf, 2017). Average panel effects can obscure that heterogeneity.

4.6 Limitations

1. **Observational identification.** Fixed effects remove time-invariant confounders and year shocks, not unobserved country-specific changes. None of the models constitutes a natural experiment.
2. **Measurement.** Both treatment and outcomes are expert-coded. Shared perceptions, retrospective coding, and reduced resolution in some countries can induce correlated error.
3. **Forward-fill assumption.** Populism is held constant between elections. Annual sample size overstates the number of distinct political changes, and election-only estimates are less precise.
4. **Persistence and dynamics.** Near-unit outcome persistence complicates level-based Granger tests. Dynamic fixed effects may be biased in finite panels, while first differences can be noisy.
5. **Sample selection.** Strict coder filtering improves reliability but reduces coverage from 163 to 96 countries. Legislative missingness is linked to the existence of legislatures.
6. **Temporal scope.** The analysis ends in 2019 and cannot evaluate the most recent wave of populist governments.
7. **Multiplicity and dependence.** FDR correction limits false discoveries, but corruption dimensions are correlated and the legislative signal remains one result among several related specifications.
8. **Average effects.** Country-level averages cannot identify heterogeneous effects by ideology, coalition structure, leader, or constitutional system without further design choices.

4.7 Conclusion

This study asked whether governing-party populism is associated with political corruption, which variable comes first, and which institution is most affected. A simple positive answer does not survive the analytical sequence. The pooled and between-country coefficients are positive but imprecise. The main within-country coefficient is small and negative, significant in controlled static models but not under every robustness transformation. Composite lead-lag, Granger, event-study, and first-difference evidence provides no robust temporal direction.

The institutional decomposition adds two qualifications. Judicial corruption shows the strongest negative same-year association, particularly at lower democracy levels. Legislative corruption

shows a small positive two-year signal after the outcome's own history and multiplicity are considered, but that result is not confirmed in first differences or election-only samples. The most defensible conclusion is therefore a negative and nuanced one: the data do not support a broad claim that populist government systematically raises political corruption, though a fragile delayed legislative channel merits further research.

Future work should exploit quasi-experimental changes in governing coalitions, explicitly model election timing and tenure, distinguish ideological families of populism, and combine expert indices with administrative corruption outcomes. Those extensions would be better positioned to separate institutional capture from perception, selection, and common political shocks.

Supplementary Evidence and Reproducibility

A.1 Selected robustness results

Table A.1: Robustness of the composite contemporaneous association.

Estimator	Coefficient	SE	95% CI	<i>p</i>
TWFE, country cluster	-0.0715	0.0353	$[-0.1408, -0.0023]$	0.043
TWFE, country + year cluster	-0.0715	0.0348	$[-0.1398, -0.0033]$	0.040
TWFE, Driscoll–Kraay	-0.0715	0.0152	$[-0.1013, -0.0418]$	< 0.001
First difference	-0.0243	0.0187	$[-0.0610, 0.0124]$	0.195

Table A.2: Primary temporal tests for composite corruption.

Direction and test	Estimate / statistic	<i>p</i>	Observations	Countries
Populism → corruption, lead-lag L1	$\theta_1 = -0.056$	0.065	3,845	95
Populism → corruption, lead-lag L2 lag 1	$\theta_1 = -0.048$	0.031	3,750	95
Populism → corruption, Granger L2 joint	$\chi^2 = 0.753$	0.686	3,750	95
Corruption → populism, Granger L2 joint	$\chi^2 = 2.848$	0.241	3,750	95
Future populism placebo, joint	–	0.096	3,750	95
Future corruption placebo, joint	–	0.841	3,750	95
Populist-entry post-event joint	–	0.228	3,940	95
Populist-entry pre-trend joint	–	0.983	3,940	95

A.2 Within-country information by corruption dimension

Table A.3: Coverage, within variation, and persistence of corruption outcomes.

Outcome	<i>N</i>	Missing	Within variance	Lag-1 corr.	Invariant countries
Composite	3,965	0	12.5%	0.993	2
Executive	3,965	0	17.2%	0.987	3
Public sector	3,965	0	16.0%	0.990	3
Legislative	3,770	195	12.4%	0.992	6
Judicial	3,965	0	11.1%	0.992	5

A.3 Additional heterogeneity diagnostic

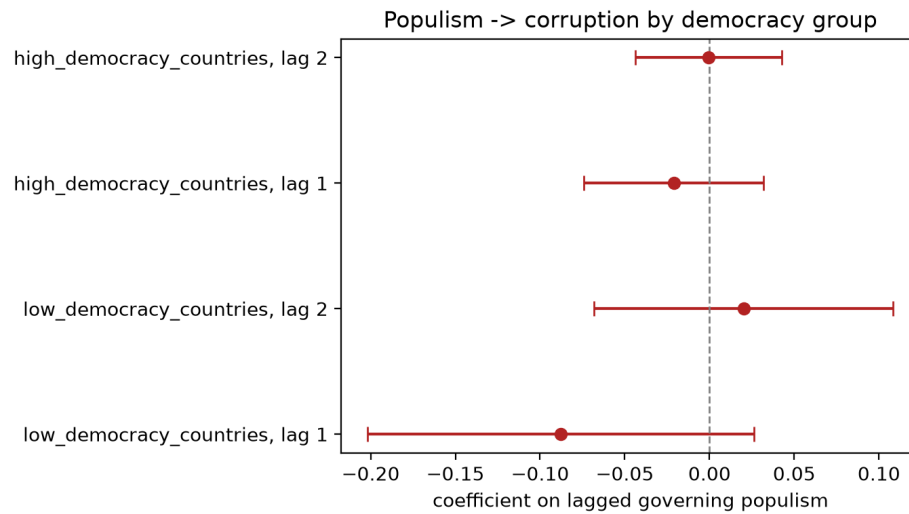


Figure A.1: Populism-to-corruption lag estimates in lower- and higher-democracy country groups. The pattern suggests more negative lag-1 coefficients at lower democracy, but marginal effects are not statistically clear.

A.4 Reproducibility workflow

The repository uses Python 3.12/3.13-compatible code managed by `uv`. Raw V-Dem and V-Party files are imported to DuckDB, while the strict processed panel is committed so all reported analyses can be reproduced without redistributing every raw file. From the repository root:

```
uv sync --extra dev --extra analysis
uv run python src/preprocessing.py
uv run pytest
uv run python scripts/generate_artifacts.py
uv run python scripts/generate_decomposition_artifacts.py
uv run python scripts/generate_report_assets.py
```

The explanatory notebooks follow the analytical sequence:

1. `01_eda_populism_corruption.ipynb`: coverage, distributions, group contrasts, and country cases;
2. `02_preprocessing.ipynb`: party aggregation, coder filtering, forward-fill, join, and validation;
3. `03_association.ipynb`: between/within comparison, TWFE, interaction, machine learning, and robustness;
4. `04_causal_direction.ipynb`: lead-lag, Granger-style tests, placebos, event studies, first differences, and electoral robustness;
5. `05_corruption_decomposition.ipynb`: dimension-specific association, temporal direction, FDR correction, and measurement checks.

The automated suite contains 33 passing tests. It checks preprocessing invariants, country-bounded lags, duplicate keys, fixed sample construction, outcome orientation, synthetic signal recovery, joint tests, multiplicity correction, and robustness coverage. All report figures are generated from the committed panel or copied from reproducible analysis artifacts; no numerical result is transcribed from an unversioned spreadsheet.

A.5 Interpretation guide

Table A.4: What each reported design can and cannot establish.

Design	Supports	Does not establish
Pooled / between OLS	Cross-country descriptive association	Within-country effect or direction
Two-way fixed effects	Same-year within-country co-movement after absorbed effects	Causal order or removal of time-varying confounding
Lead-lag FE	Temporal precedence in observed annual data	Incremental prediction beyond own persistence
Granger-style FE	Incremental predictive content of lagged proposed cause	Structural or intervention-based causality
Event study	Dynamic pattern around defined entries/shocks	Exogenous treatment assignment
First differences	Association among annual changes	Long-run level relationship
XGBoost + SHAP	Held-out-country prediction and fitted feature use	Statistical significance or causal importance

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